MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MULTIPLEXER, ANALOG, KEY CODE RRB

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic multiplexer module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.960 failures per million operating hours for class I or 8.672 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be RRB.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/321-1 for class I and M28787/321-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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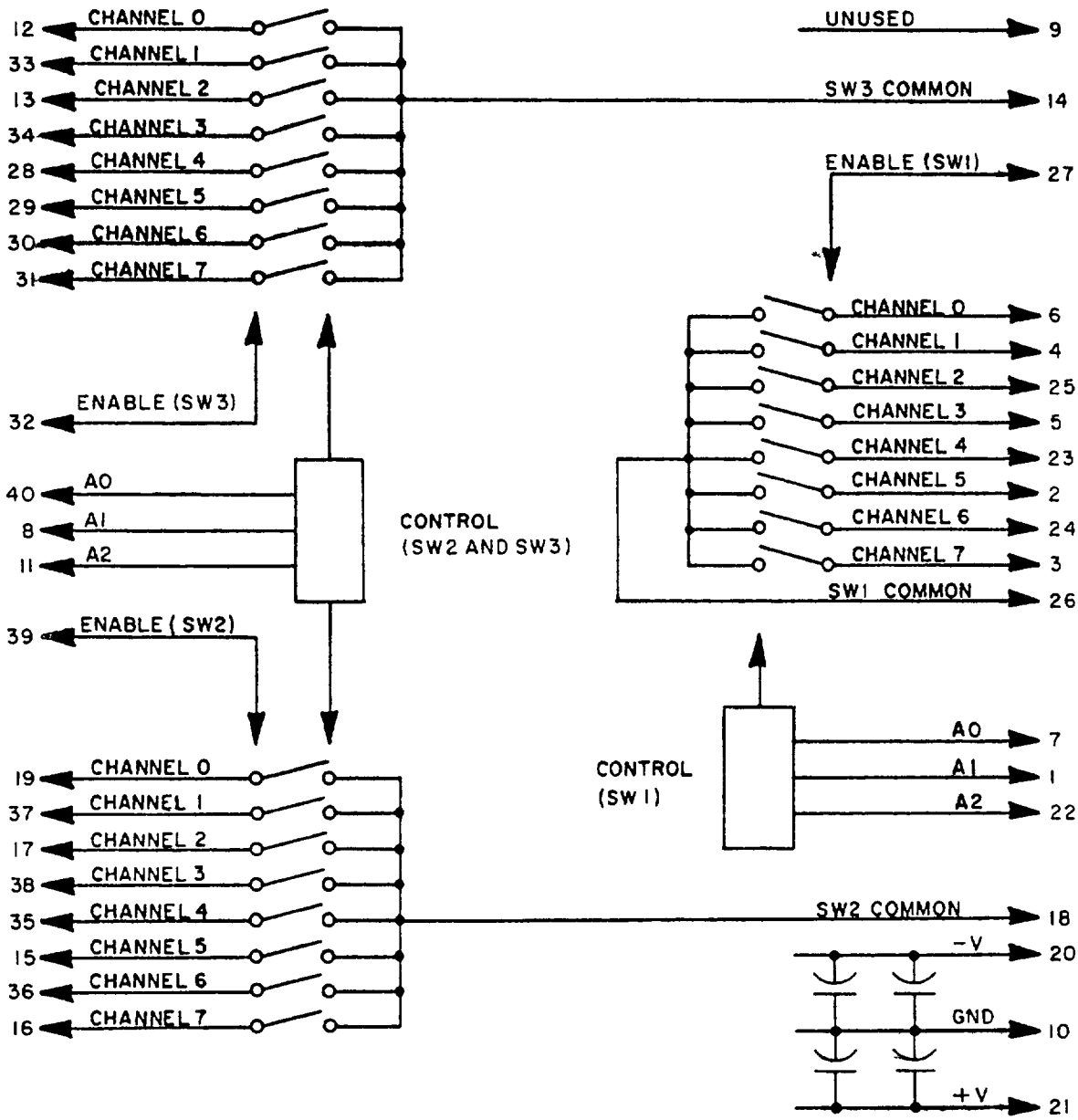


FIGURE 1. RRB module function diagram, multiplexer.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1, A2	Overvoltage check	III	Q	70	180	70	180	70	180	70	180	70	180	mA
A3 - A18	Overvoltage check	III	Q	35	55	35	55	35	55	35	55	35	55	mA
B1	Isolation													
	Initial	IV	X	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	IV	X	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B2	Isolation													
	Initial	IV	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	IV	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
C1-C24	ON resistance check													
	Initial	V	X	300	1500	100	1500	300	1700	100	1400	300	1800	Ω
	End-of-life	V	X	300	1600	100	1700	300	1900	100	1600	300	2200	Ω
D1-D16	Channel leakage current	VI	X	-50	50	-50	50	-500	500	-50	50	-750	750	nA
E1-E9	Digital input current, high level	VII	X	-1	1	-1	1	-5	5	-1	1	-5	5	µA
F1-F9	Digital input current, low level	VIII	X	-1	1	-1	1	-5	5	-1	1	-5	5	µA
G1-G24	Functional	IX	X	3.2	4.6	3.1	5.0	3.0	4.6	3.2	5.0	2.8	4.6	mA
G25, G26	Functional	IX	X	-50	50	-50	50	-500	500	-50	50	-750	750	nA
H1-H24	High frequency response check	X	S	9.67	10.40	9.67	10.50	9.59	10.40	9.72	10.50	9.54	10.40	V rms
J1	Quiescent power supply current	XI	S	0.0	6.6	0.0	6.6	0.0	6.6	0.0	6.6	0.0	6.6	mA
J2	Quiescent power supply current	XI	S	0.0	3.3	0.0	3.3	0.0	3.3	0.0	3.3	0.0	3.3	mA

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued..

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
K1, K2	Power supply transient amplitude	XII	Q	0	350	0	350	0	350	0	350	0	350	mV
L1-L27	Switching time (t_{ON})	XIII	S	0.03	1.00	0.03	1.00	0.03	1.00	0.03	1.00	0.03	1.00	µs
M1-M27	Switching time (t_{OFF})	XIV	S	0.03	0.80	0.03	0.80	0.03	0.80	0.03	0.80	0.03	0.80	µs
N1-N6	Break-before-make delay	XV	S	30	500	30	500	30	500	30	500	30	500	ns
P1-P16	Power supply rejection ratio check	XVI	S	-∞	-60	-∞	-60	-∞	-60	-∞	-60	-∞	-60	dB
Q1-Q3	Output crosstalk check	XVII	Q	-∞	-30	-∞	-30	-∞	-30	-∞	-30	-∞	-30	dB
Q4-Q30	Output crosstalk check	XVII	Q	-∞	-50	-∞	-50	-∞	-50	-∞	-50	-∞	-50	dB
R1, R2	Power supply filter capacitance	XVIII	X	0.75	1.25	0.75	1.25	0.75	1.25	0.75	1.25	0.75	1.25	µF
S1, S3	Input capacitance	XVIII	Q	2	30	2	30	2	30	2	30	2	30	pF
S4-S33	Input capacitance	XVIII	Q	2	15	2	15	2	15	2	15	2	15	pF
T1-T3	Output capacitance	XVIII	Q	2	50	2	50	2	50	2	50	2	50	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows: A, B where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C and G through R shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2. Test conditions D, E, and F shall be performed at 25°C, 60°C, and 85°C using the initial limits for the applicable temperature.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 3. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through R shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests B through T shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

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* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation	555	555	mW
Supply voltages (see 6.5.1)	±18	±18	V
Input levels:			
Digital			
Low level	-V-4 min, 0.8 max	-V-4 min, 0.8 max	V
High level	4 min, +V+4 max	4 min, +V+4 max	V
Analog	±18	±18	V

6.5 Module function. This module contains three 8 input analog multiplexers controlled by six digital inputs. Two multiplexers are controlled by three digital bits paralleled to both multiplexers. The third multiplexer has its own three bit digital control lines. In addition, each multiplexer has its own inhibit line.

6.5.1 Maximum ratings. The maximum ratings for supply voltages and input levels given in 6.4 represent extremes beyond which damage to the module may occur. Operation at the extremes is performed in this specification to ensure that the module will survive these levels without damage. Performance specifications are tested at supply levels 8 percent less than maximum ratings and digital and analog input levels equal to or between the supply voltages.

6.5.2 Module handling procedures. This module contains CMOS devices which are susceptible to damage when exposed to a static electrical charge. When handling, shipping, or storing, all module pins should be shorted or the module should be placed in an electrically conductive package. Soldering iron tips and all test equipment must be grounded. Failure to comply with these guidelines could cause irreversible module damage.

6.5.3 Input to output relationship. The digital input versus ON channel relationship is shown in table II.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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TABLE II. Input to output relationship. 1/-3/

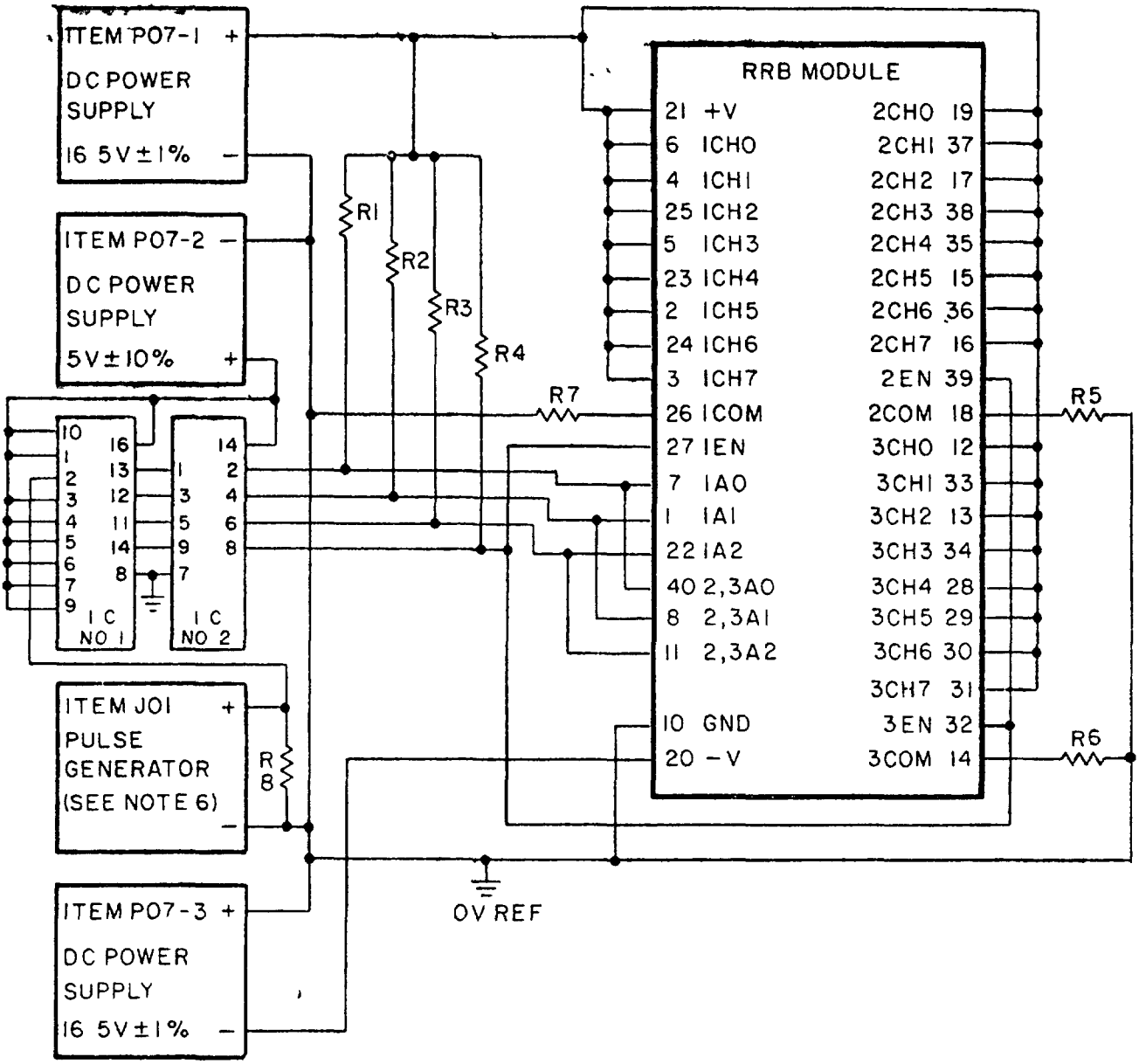
Enable 27	A2 22	A1 1	A0 7	Channel ON	Enable 32	A2 11	A1 8	A0 40	Channel ON	Enable 39	A2 11	A1 8	A0 40	Channel ON
H	L	L	L	6←----→26	H	L	L	L	12←------→14	H	L	L	L	19←-----→18
H	L	L	H	4←----→26	H	L	L	H	33←------→14	H	L	L	H	37←-----→18
H	L	H	L	25←----→26	H	L	H	L	13←------→14	H	L	H	L	17←-----→18
H	L	H	H	5←----→26	H	L	H	H	34←------→14	H	L	H	H	38←-----→18
H	H	L	L	23←----→26	H	H	L	L	28←------→14	H	H	L	L	35←-----→18
H	H	L	H	2←----→26	H	H	L	H	29←------→14	H	H	L	H	15←-----→18
H	H	H	L	24←----→26	H	H	H	L	30←------→14	H	H	H	L	36←-----→18
H	H	H	H	3←----→26	H	H	H	H	31←------→14	H	H	H	H	16←-----→18
L	X	X	X	None	L	X	X	X	None	L	X	X	X	None

1/ L = Low-level = GND.

2/ H = High-level = +V.

3/ X = Don't care condition.

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NOTES:
1. R1-R4 = 1 kΩ ±1%, 1/2 W.
2. R5-R7 = 604Ω ±1%, 1/2 W.
3. R8 = 51.1Ω ±5%, 1 W.
4. IC No. 1 = Fairchild 9316P, or equivalent.
5. IC No. 2 = Texas Instruments SN5407N, or equivalent.
6. Item J01 setup.
   Up level = 4.8 ±0.2 V.
   Down level = 0.4 ±0.4 V.
   Duty cycle = 50 ±5%.
   $t_{THL} = t_{TLH} = 30 \pm 25$ ns.
   Frequency = 100 kHz ±5%.

FIGURE 2. Operating temperature, component temperature, and life test circuit.

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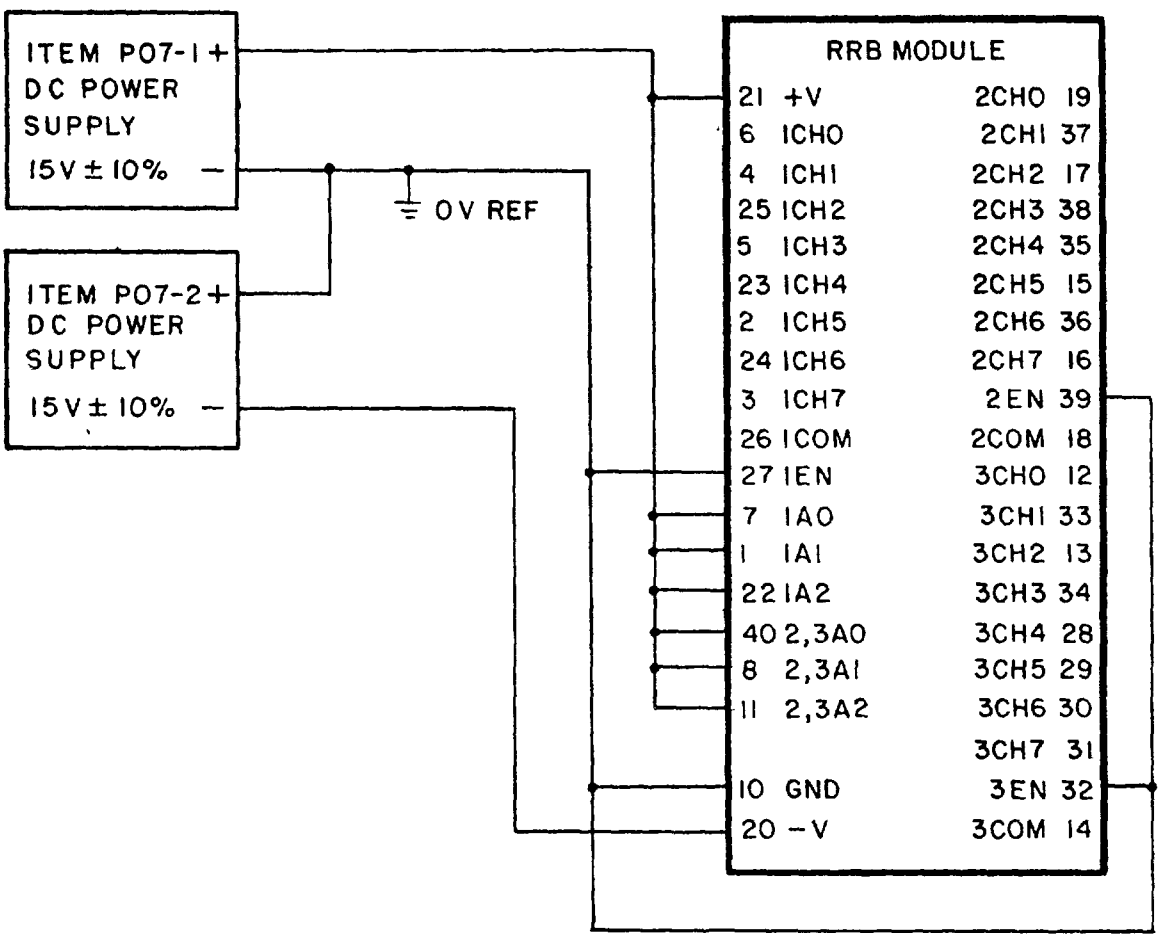


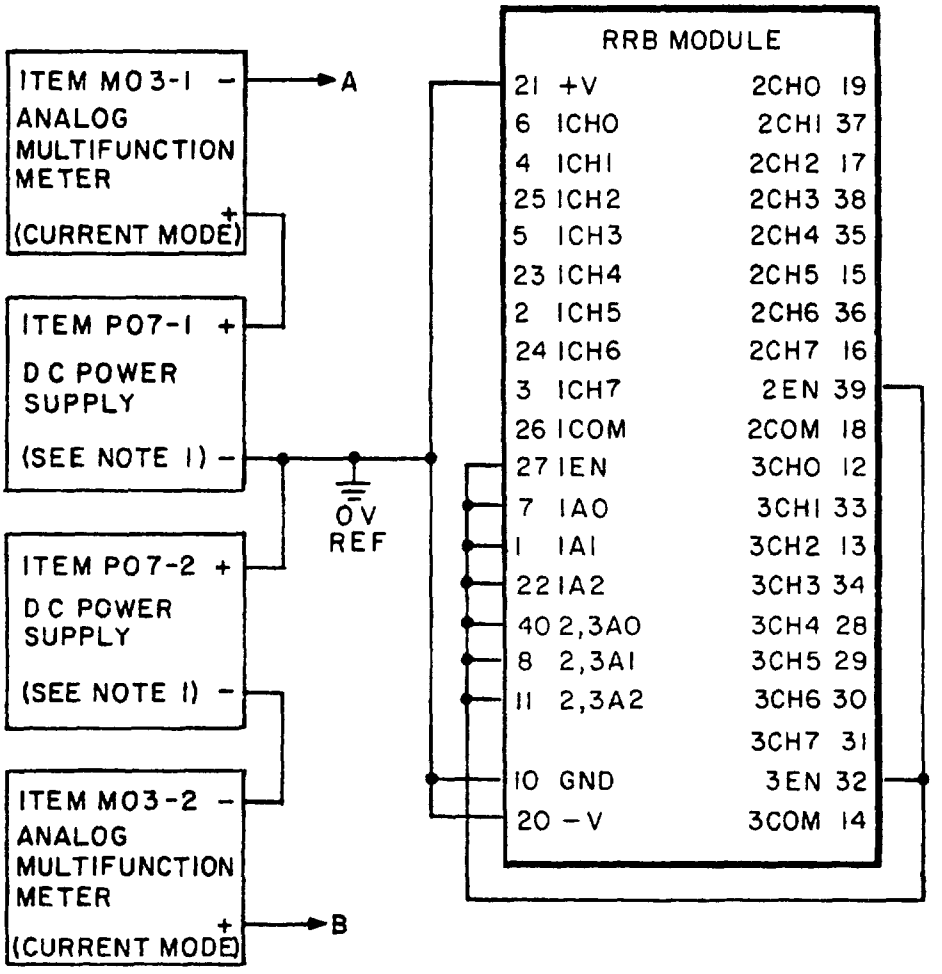
FIGURE 3. Vibration test curcuit.

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* TABLE III. Overvoltage check test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Overvoltage check	27	Open	70	180	mA	4
A2	"	Open	27	70	180	"	"
A3	"	6, 12, 19	Open	35	55	"	"
A4	"	23, 28, 35	"	"	"	"	"
A5	"	4, 33, 37	"	"	"	"	"
A6	"	2, 15, 29	"	"	"	"	"
A7	"	13, 17, 25	"	"	"	"	"
A8	"	24, 30, 36	"	"	"	"	"
A9	"	5, 34, 38	"	"	"	"	"
A10	"	3, 16, 31	"	"	"	"	"
A11	"	Open	6, 12, 19	"	"	"	"
A12	"	"	23, 28, 35	"	"	"	"
A13	"	"	4, 33, 37	"	"	"	"
A14	"	"	2, 15, 29	"	"	"	"
A15	"	"	13, 17, 25	"	"	"	"
A16	"	"	24, 30, 36	"	"	"	"
A17	"	"	5, 34, 38	"	"	"	"
A18	"	"	3, 16, 31	"	"	"	"

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NOTES:
1. Set items PO7-1 and PO7-2 to 4 V ±1% for test conditions A1 and A2 and to 18 V ±1% for test conditions A3-A18.
2. Since items PO7-1 and PO7-2, MO3-1 and MO3-2 are not used simultaneously, only one item PO7 and one item MO3 is required.

FIGURE 4. Overvoltage check test circuit for table III.

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TABLE IV. Isolation test.

Test conndition	Parameter	Connect item R06 1/		Initial limits 25°C ±5°C ambient		Unit
		+terminal to pin	-terminal to pin	Min	Max	
B1	Isolation	All pins except pin 9	9	10	∞	MΩ
B2	Isolation	All pins	Frame 2/	10	∞	MΩ

1/ Item R06 setup:
V = 100 V dc ±10%.

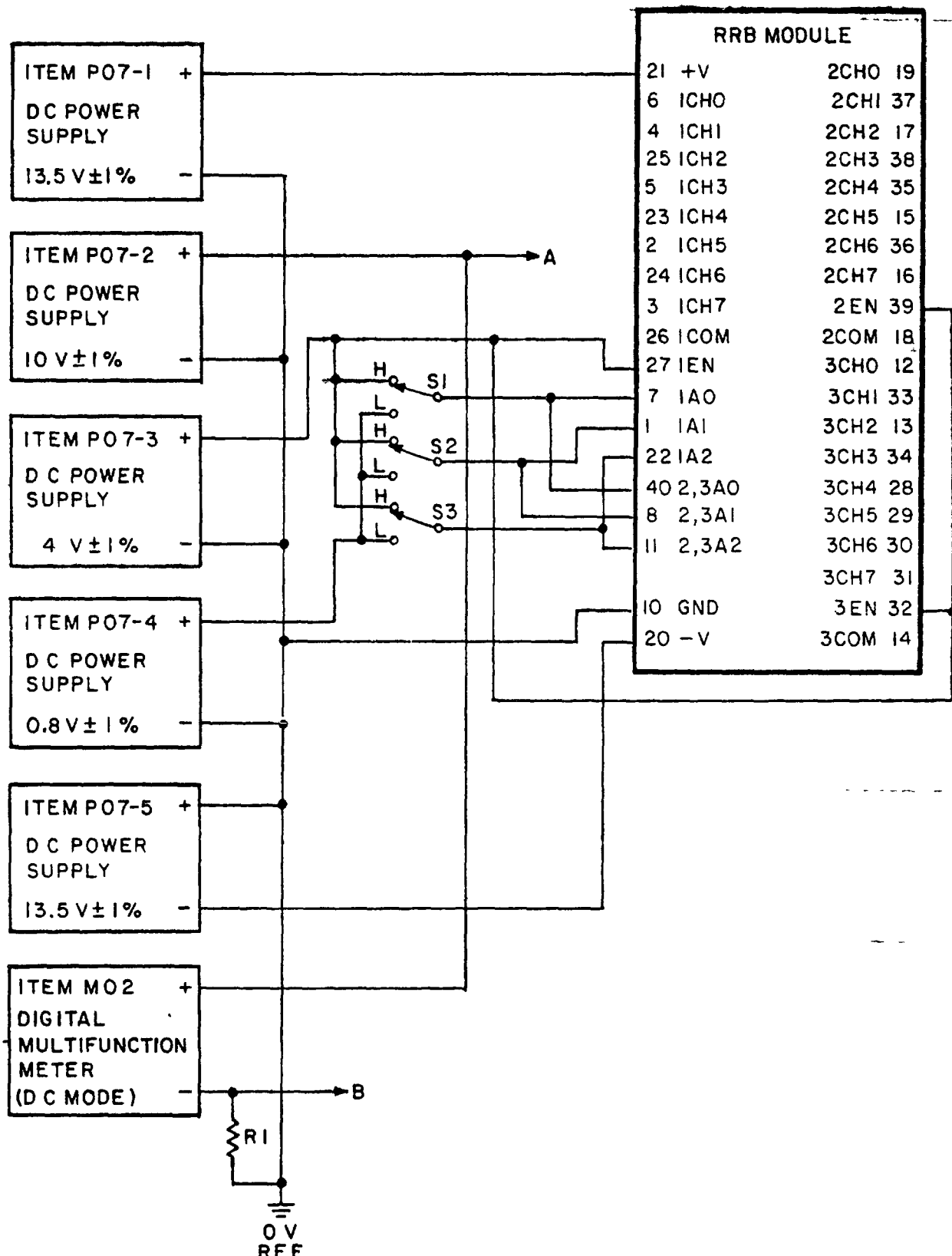
2/ On test condition B2 (when measuring to frame) penetrate the anodized finish of the module frame with a sharply pointed probe connected to the negative terminal.

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TABLE V. ON resistance check test.

Test condition	Parameter	Switch position			Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		S3	S2	S1	A to pin	B to pin	Min	Max		
C1	1CH0 ON resistance check 1/	L	L	L	6	26	30	150	mV	5
C2	1CH1 "	L	L	H	4	"	"	"	"	"
C3	1CH2 "	L	H	L	25	"	"	"	"	"
C4	1CH3 "	L	H	H	5	"	"	"	"	"
C5	1CH4 "	H	L	L	23	"	"	"	"	"
C6	1CH5 "	H	L	H	2	"	"	"	"	"
C7	1CH6 "	H	H	L	24	"	"	"	"	"
C8	1CH7 "	H	H	H	3	"	"	"	"	"
C9	2CH0 "	L	L	L	19	18	"	"	"	"
C10	2CH1 "	L	L	H	37	"	"	"	"	"
C11	2CH2 "	L	H	L	17	"	"	"	"	"
C12	2CH3 "	L	H	H	38	"	"	"	"	"
C13	2CH4 "	H	L	L	35	"	"	"	"	"
C14	2CH5 "	H	L	H	15	"	"	"	"	"
C15	2CH6 "	H	H	L	36	"	"	"	"	"
C16	2CH7 "	H	H	H	16	"	"	"	"	"
C17	3CH0 "	L	L	L	12	14	"	"	"	"
C18	3CH1 "	L	L	H	33	"	"	"	"	"
C19	3CH2 "	L	H	L	13	"	"	"	"	"
C20	3CH3 "	L	H	H	34	"	"	"	"	"
C21	3CH4 "	H	L	L	28	"	"	"	"	"
C22	3CH5 "	H	L	H	29	"	"	"	"	"
C23	3CH6 "	H	H	L	30	"	"	"	"	"
C24	3CH7 "	H	H	H	31	"	"	"	"	"

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NOTES:
1. R1 = 100 kΩ ±1%, 1/8 W.
2. S1 = S2 = S3 = SPDT switch.

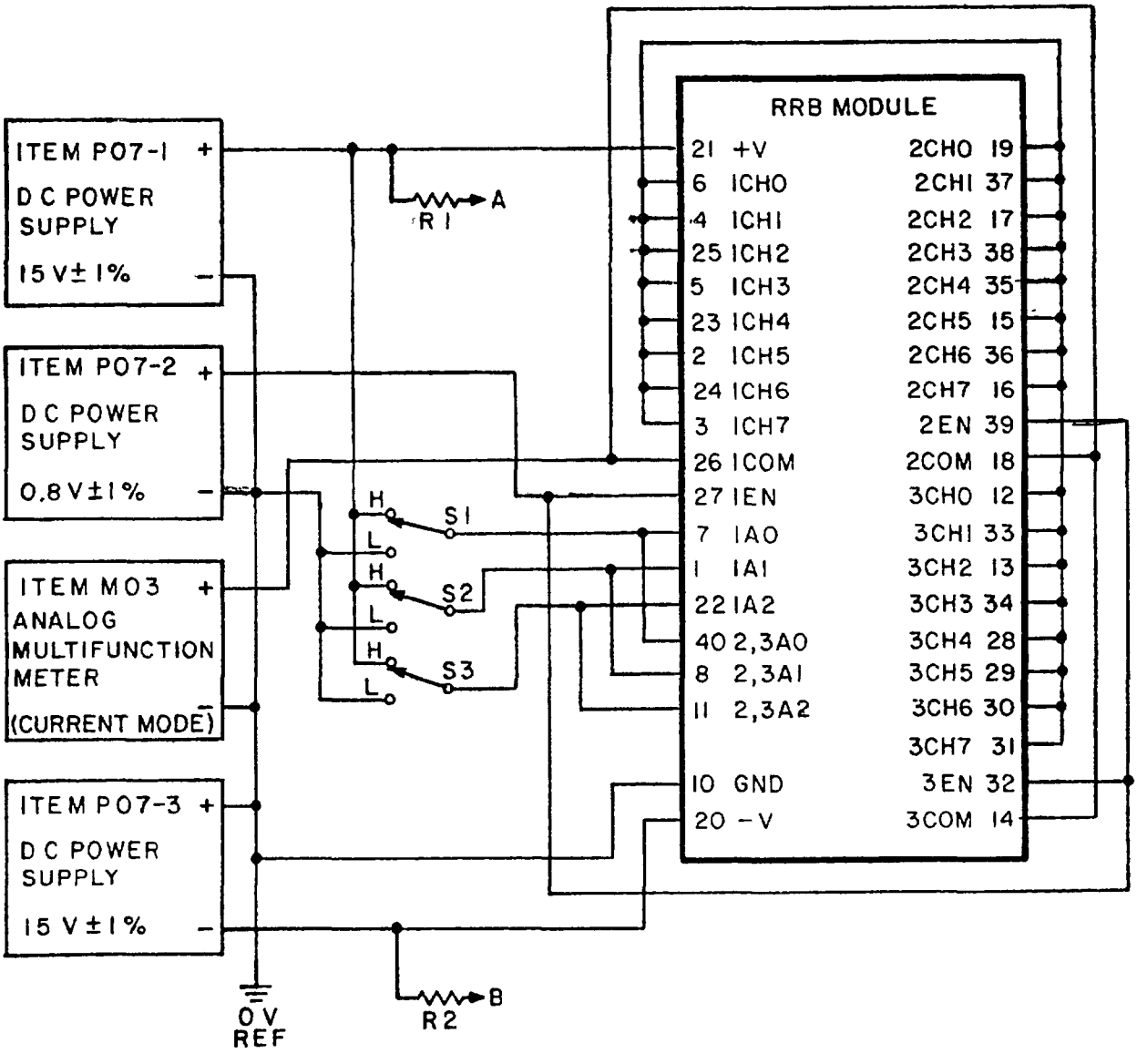
FIGURE 5. ON resistance check test circuit for table V.

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TABLE VI. Channel leakage current test.

Test condition	Parameter	Switch position			Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		S3	S2	S1	A to pin	B to pin	Min	Max		
D1	Channel leakage current	L	L	L	6	Open	-50	50	nA	6
D2	"	L	L	H	"	"	"	"	"	"
D3	"	L	H	L	"	"	"	"	"	"
D4	"	L	H	H	"	"	"	"	"	"
D5	"	H	L	L	"	"	"	"	"	"
D6	"	H	L	H	"	"	"	"	"	"
D7	"	H	H	L	"	"	"	"	"	"
D8	"	H	H	H	"	"	"	"	"	"
D9	"	L	L	L	Open	6	"	"	"	"
D10	"	L	L	H	"	"	"	"	"	"
D11	"	L	H	L	"	"	"	"	"	"
D12	"	L	H	H	"	"	"	"	"	"
D13	"	H	L	L	"	"	"	"	"	"
D14	"	H	L	H	"	"	"	"	"	"
D15	"	H	H	L	"	"	"	"	"	"
D16	"	H	H	H	"	"	"	"	"	"

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NOTES:
1. R1 = R2 = 604Ω ±1%, 1/2 W.
2. S1 - S3 = SPDT switch.

FIGURE 6. Channel leakage current test circuit for table VI.

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TABLE VII. Digital input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	1EN Digital input current, high level	7, 1, 22, 40, 8, 11, 39, 32	27	-1	1	µA	7
E2	1A0 "	27, 1, 22, 40, 8, 11, 39, 32	7	"	"	"	"
E3	1A1 "	27, 7 22, 40, 8, 11, 39, 32	1	"	"	"	"
E4	1A2 "	27, 7, 1, 40, 8, 11, 39, 32	22	"	"	"	"
E5	2, 3A0 "	27, 7, 1, 22, 8, 11, 39, 32	40	"	"	"	"
E6	2, 3A1 "	27, 7, 1, 22, 40, 11, 39, 32	8	"	"	"	"
E7	2, 3A2 "	27, 7, 1, 22, 40, 8, 39, 32	11	"	"	"	"
E8	2EN "	27, 7, 1, 22, 40, 8, 11, 32	39	"	"	"	"
E9	3EN "	27, 7, 1, 22, 40, 8, 11, 39	32	"	"	"	"

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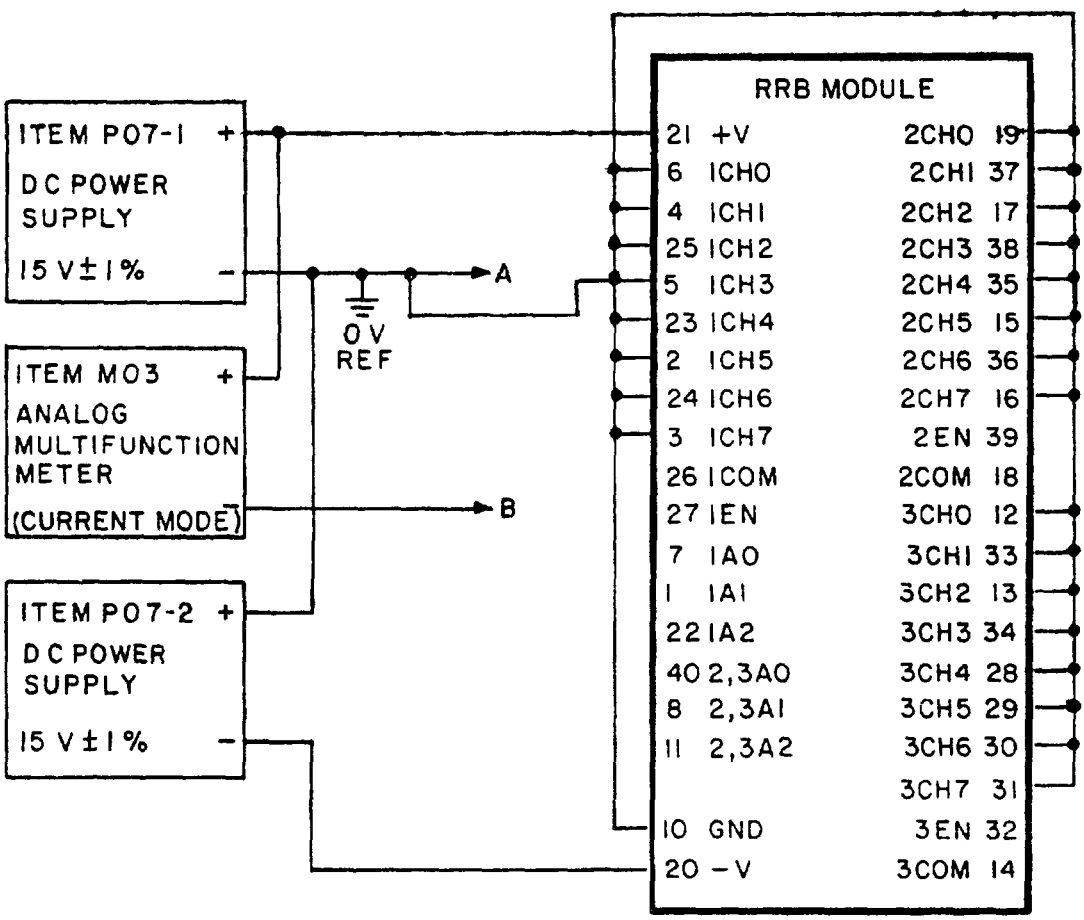


FIGURE 7. Digital input current (high-level) test circuit for table VII.

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TABLE VIII. Digital input current (low-level) test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
F1	1EN	Digital input current, low level	7, 1, 22, 40, 8, 11, 39, 32	27	-1	1	µA	8
F2	1A0	"	27, 1, 22, 40, 8, 11, 39, 32	7	"	"	"	"
F3	1A1	"	27, 7, 22, 40, 8, 11, 39, 32	1	"	"	"	"
F4	1A2	"	27, 7, 1, 40, 8, 11, 39, 32	22	"	"	"	"
F5	2, 3A0	"	27, 7, 1, 22, 8, 11, 39, 32	40	"	"	"	"
F6	2, 3A1	"	27, 7, 1, 22, 40, 11, 39, 32	8	"	"	"	"
F7	2, 3A2	"	27, 7, 1, 22, 40, 8, 39, 32	11	"	"	"	"
F8	2EN	"	27, 7, 1, 22, 40, 8, 11, 32	39	"	"	"	"
F9	3EN	"	27, 7, 1, 22, 40, 8, 11, 39	32	"	"	"	"

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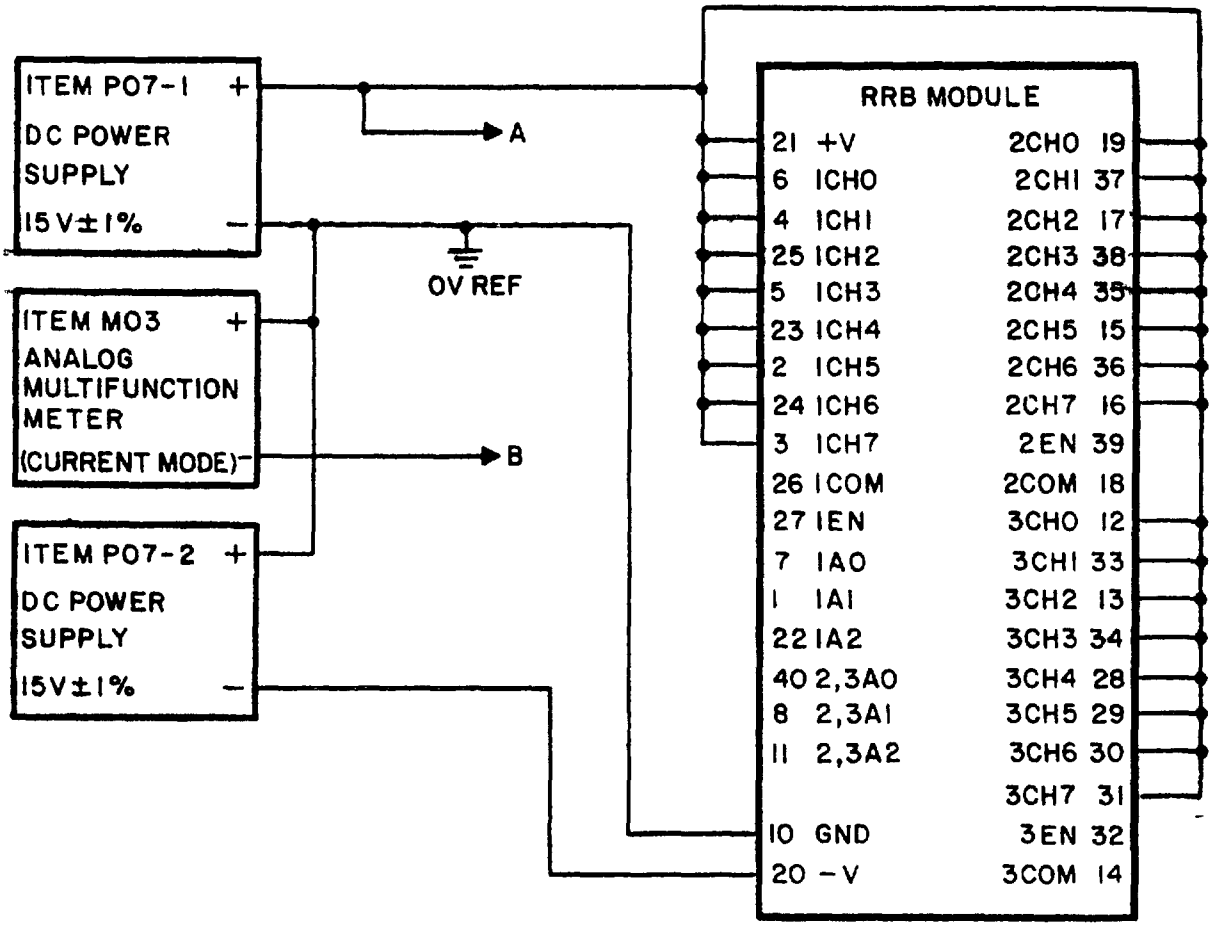


FIGURE 8. Digital input current (low-level) test circuit for table VIII.

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TABLE IX. Functional test.

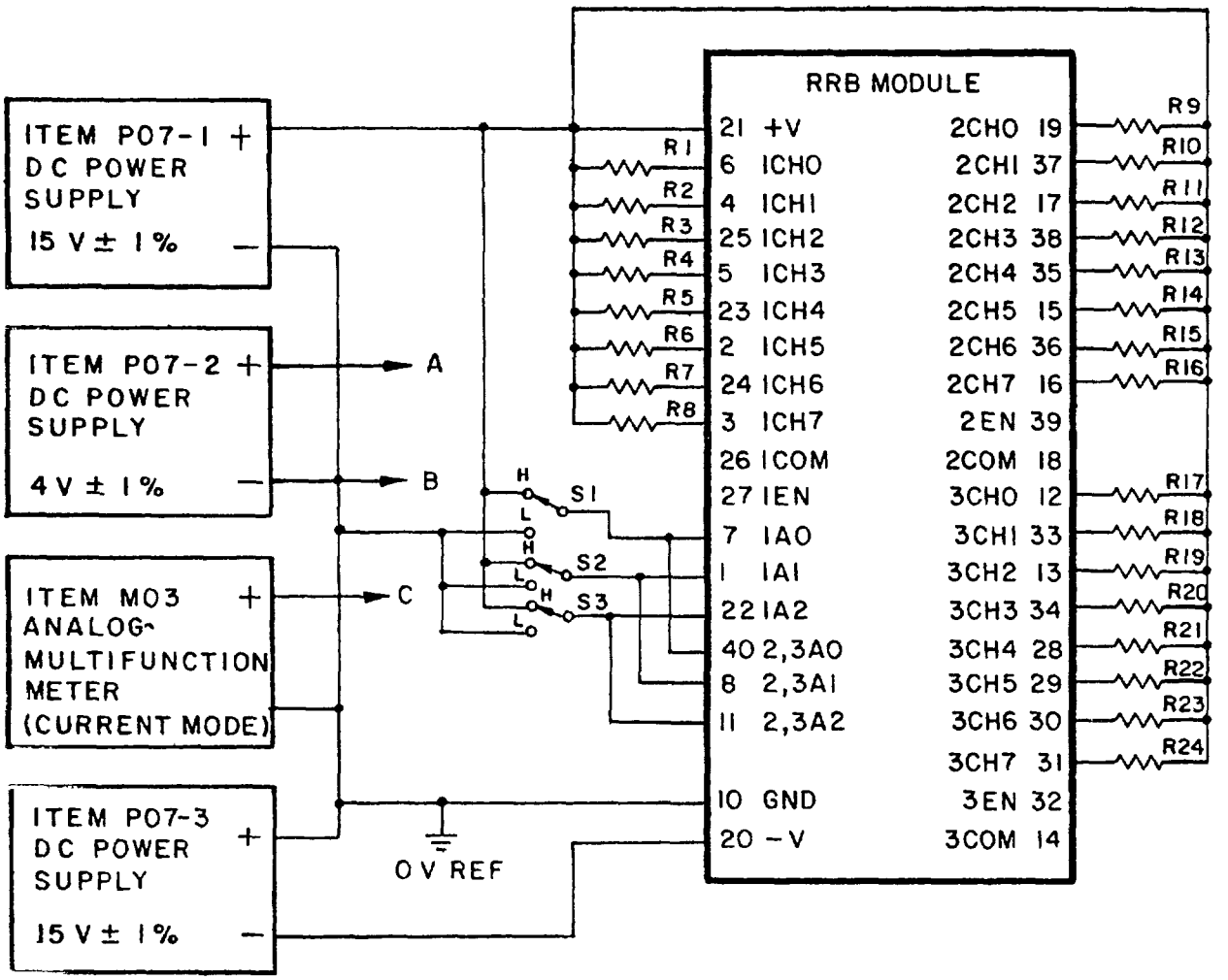
Test condition	Parameter	Switch position			Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		S3	S2	S1	A to pin	B to pin	C to pin	Min	Max		
G1	Functional	L	L	L	27	39, 32	14, 18, 26	3.2	4.6	mA	9
G2	"	L	L	H	"	"	"	"	"	"	"
G3	"	L	H	L	"	"	"	"	"	"	"
G4	"	L	H	H	"	"	"	"	"	"	"
G5	"	H	L	L	"	"	"	"	"	"	"
G6	"	H	L	H	"	"	"	"	"	"	"
G7	"	H	H	L	"	"	"	"	"	"	"
G8	"	H	H	H	"	"	"	"	"	"	"
G9	"	L	L	L	39	27, 32	"	"	"	"	"
G10	"	L	L	H	"	"	"	"	"	"	"
G11	"	L	H	L	"	"	"	"	"	"	"
G12	"	L	H	H	"	"	"	"	"	"	"
G13	"	H	L	L	"	"	"	"	"	"	"
G14	"	H	L	H	"	"	"	"	"	"	"
G15	"	H	H	L	"	"	"	"	"	"	"
G16	"	H	H	H	"	"	"	"	"	"	"
G17	"	L	L	L	32	27, 39	"	"	"	"	"
G18	"	L	L	H	"	"	"	"	"	"	"
G19	"	L	H	L	"	"	"	"	"	"	"
G20	"	L	H	H	"	"	"	"	"	"	"

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TABLE IX. Functional test - Continued.

Test condition	Parameter	Switch position			Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		S3	S2	S1	A to pin	B to pin	C to pin	Min	Max		
G21	Functional	H	L	L	32	27, 39	14, 18, 26	3.2	4.6	mA	9
G22	"	H	L	H	"	"	"	"	"	"	"
G23	"	H	H	L	"	"	"	"	"	"	"
G24	"	H	H	H	"	"	"	"	"	"	"
G25	"	H	H	H	"	"	18, 26	-50	50	nA	"
G26	"	H	H	H	39	27, 32	26	-50	50	nA	"

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NOTES:
1. R1 - R24 = 3 kΩ ±1%, 1/8 W.
2. S1 - S3 = SPDT switch.

FIGURE 9. Functional test curcuit for table IX.

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TABLE X. High Frequency response check test.

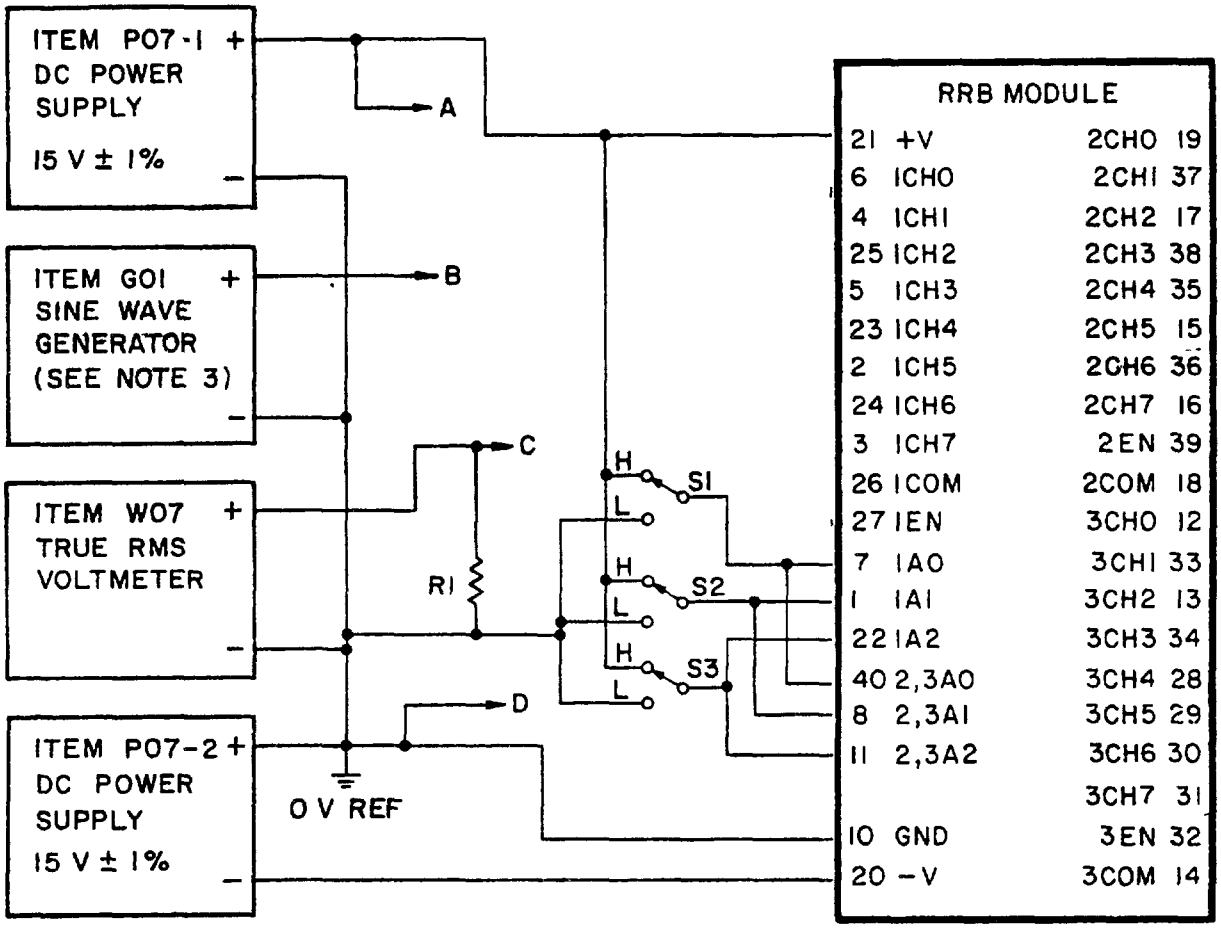
Test condition	Parameter	Switch position			Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		S3	S2	S1	A to pin	B to pin	C to pin	D to pin	Min	Max		
H1	High frequency response check	L	L	L	27	6	26	32, 39	9.67	10.40	V rms	10
H2	"	L	L	H	"	4	"	"	"	"	"	"
H3	"	L	H	L	"	25	"	"	"	"	"	"
H4	"	L	H	H	"	5	"	"	"	"	"	"
H5	"	H	L	L	"	23	"	"	"	"	"	"
H6	"	H	L	H	"	2	"	"	"	"	"	"
H7	"	H	H	L	"	24	"	"	"	"	"	"
H8	"	H	H	H	"	3	"	"	"	"	"	"
H9	"	L	L	L	39	19	18	27, 32	"	"	"	"
H10	"	L	L	H	"	37	"	"	"	"	"	"
H11	"	L	H	L	"	17	"	"	"	"	"	"
H12	"	L	H	H	"	38	"	"	"	"	"	"
H13	"	H	L	L	"	35	"	"	"	"	"	"
H14	"	H	L	H	"	15	"	"	"	"	"	"
H15	"	H	H	L	"	36	"	"	"	"	"	"
H16	"	H	H	H	"	16	"	"	"	"	"	"
H17	"	L	L	L	32	12	14	27, 39	"	"	"	"

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TABLE X. High frequency response check test - Continued.

Test condition	Parameter	Switch position			Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		S3	S2	S1	A to pin	B to pin	C to pin	D to pin	Min	Max		
H18	High frequency response check	L	L	H	32	33	14	27, 39	9.67	10.40	V rms	10
H19	"	L	H	L	"	13	"	"	"	"	"	"
H20	"	L	H	H	"	34	"	"	"	"	"	"
H21	"	H	L	L	"	28	"	"	"	"	"	"
H22	"	H	L	H	"	29	"	"	"	"	"	"
H23	"	H	H	L	"	30	"	"	"	"	"	"
H24	"	H	H	H	"	31	"	"	"	"	"	"

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NOTES:
1. R1 = 20 kΩ ±1%, 1/8 W.
2. S1-S3 = SPDT switch.
3. Item G01 setup:
   Frequency = 100 kHz ±1%.
   Sine wave amplitude = 10.5 V rms ±1%.

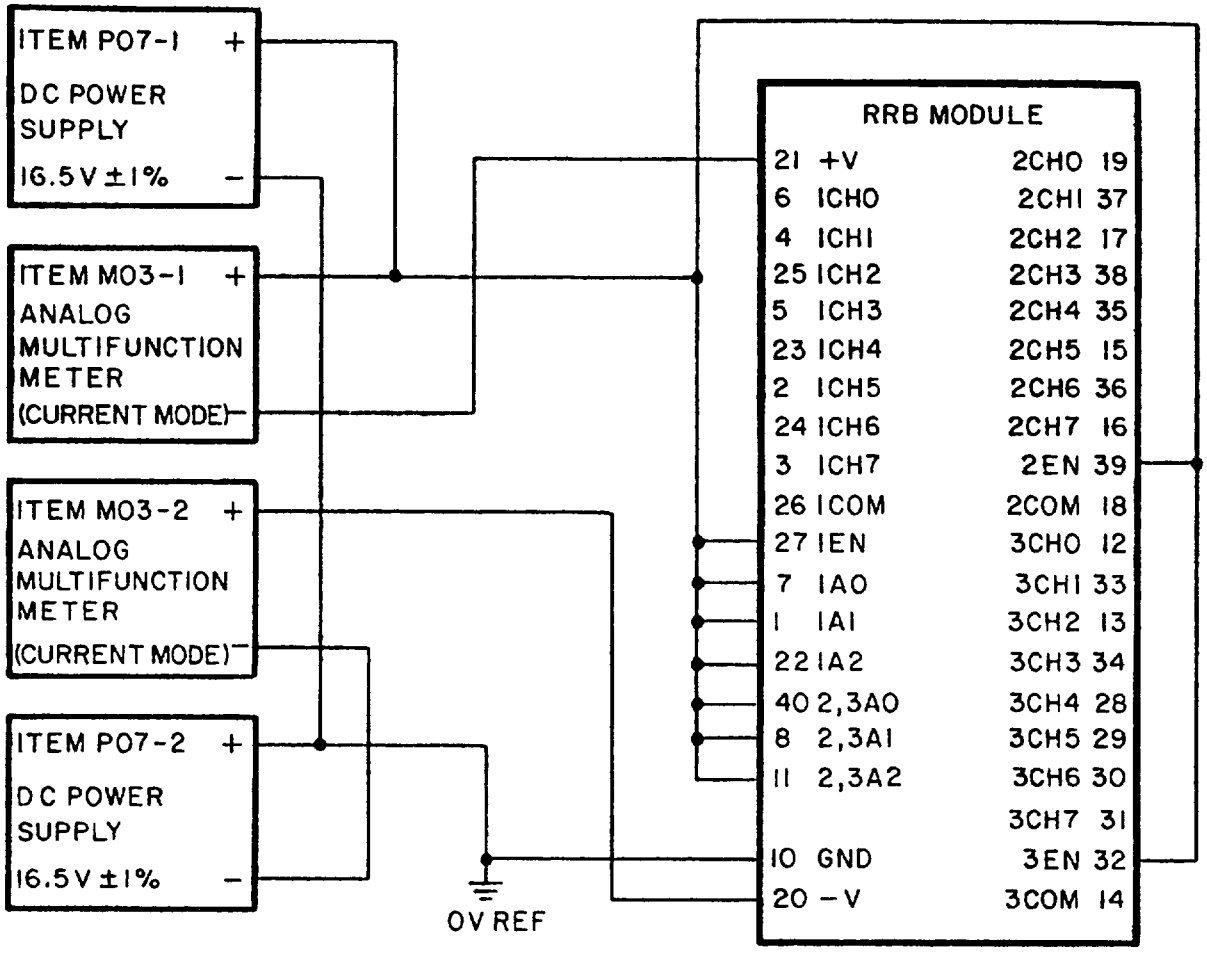
FIGURE 10. High frequency response check test circuit for table X.

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TABLE XI. Quiescent power supply current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
J1	Quiescent power supply current	0.0	6.6	mA	11
J2	Quiescent power supply current	0.0	3.3	mA	11

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NOTES:
1. Since items M03-1 and M03-2 are not used simultaneously, only one item M03 is required.
2. Read item M03-1 for test condition J1 and item M03-2 for test condition J2.
3. Measurements shall be conducted after power has been applied a minimum of 4 minutes.

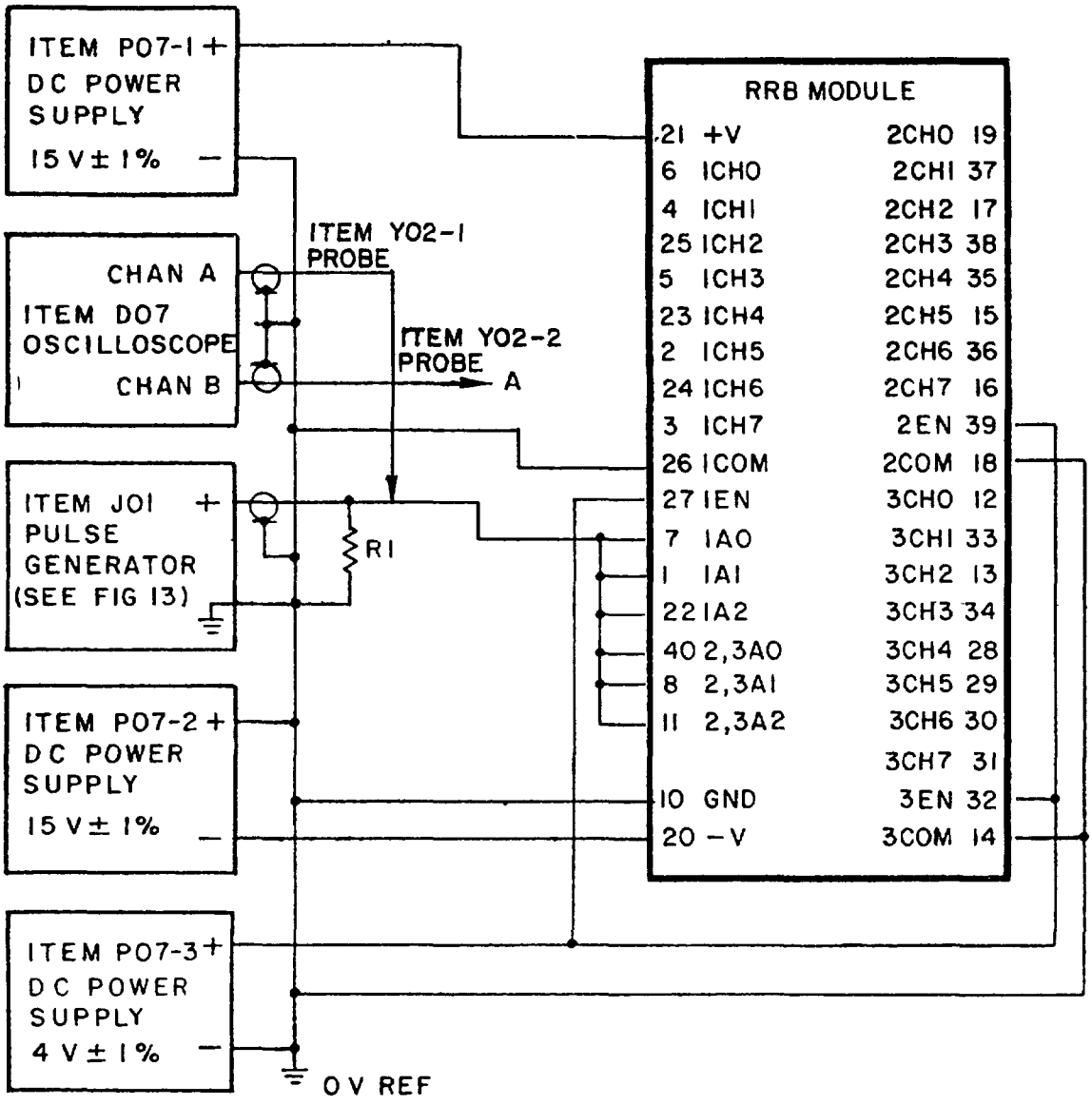
FIGURE 11. Quiescent power supply current test circuit for table XI.

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TABLE XII. Power supply transient amplitude test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
K1	Power supply transient amplitude	21	0	350	mV	12, 13
K2	Power supply transient amplitude	20	0	350	mV	12, 13

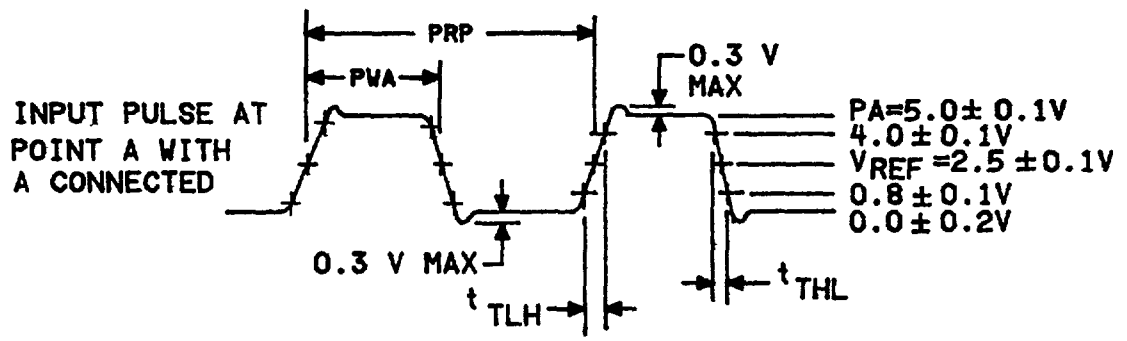
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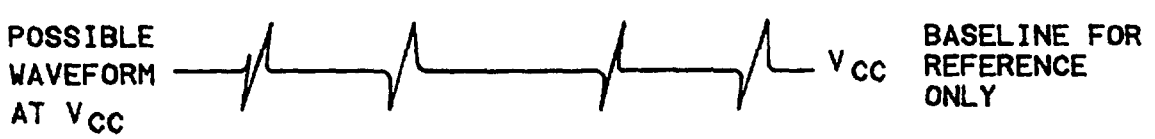


NOTE: R1 = 51.1Ω ±5%, 1 W.

FIGURE 12. Power supply transient amplitude test circuit for table XII.

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NOTES:
1. Input pulse parameters:
   PWA = 5.0 ±0.5 µs.
   PRP = 10 ±1 µs.
   $t_{TLH} = t_{THL}$ = 10 ±1 ns (verified at each input pin).
2. For test condition K1 read most negative peak at pin 21 reference to baseline.
3. For test condition K2 read most negative peak at pin 20 reference to baseline.

FIGURE 13. Power supply transient amplitude test circuit waveforms for table XII.

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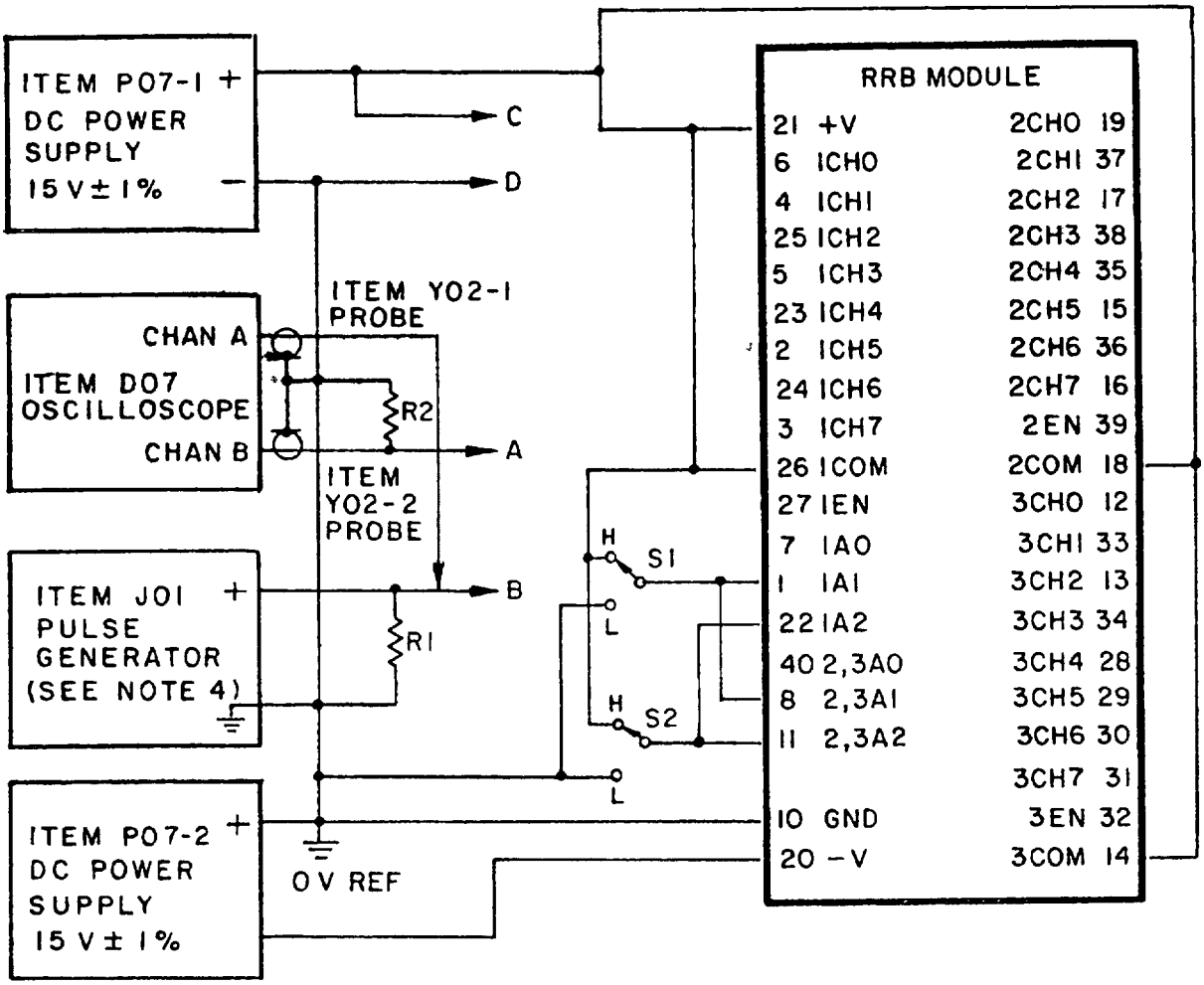
Test condition	Parameter		Switch position S2	Switch position S1	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
L1	1CH0	Switching time (t_{ON})	L	L	6	7, 40	27, 32, 39	Open	0.03	1.00	µs	14, 15
L2	1CH1	"	L	L	4	"	"	"	"	"	"	"
L3	1CH2	"	L	H	25	"	"	"	"	"	"	"
L4	1CH3	"	L	H	5	"	"	"	"	"	"	"
L5	1CH4	"	H	L	23	"	"	"	"	"	"	"
L6	1CH5	"	H	L	2	"	"	"	"	"	"	"
L7	1CH6	"	H	H	24	"	"	"	"	"	"	"
L8	1CH7	"	H	H	3	"	"	"	"	"	"	"
L9	2CH0	"	L	L	19	"	"	"	"	"	"	"
L10	2CH1	"	L	L	37	"	"	"	"	"	"	"
L11	2CH2	"	L	H	17	"	"	"	"	"	"	"
L12	2CH3	"	L	H	38	"	"	"	"	"	"	"
L13	2CH4	"	H	L	35	"	"	"	"	"	"	"
L14	2CH5	"	H	L	15	"	"	"	"	"	"	"
L15	2CH6	"	H	H	36	"	"	"	"	"	"	"
L16	2CH7	"	H	H	16	"	"	"	"	"	"	"
L17	3CH0	"	L	L	12	"	"	"	"	"	"	"
L18	3CH1	"	L	L	33	"	"	"	"	"	"	"
L19	3CH2	"	L	H	13	"	"	"	"	"	"	"
L20	3CH3	"	L	H	34	"	"	"	"	"	"	"
L21	3CH4	"	H	L	28	"	"	"	"	"	"	"
L22	3CH5	"	H	L	29	"	"	"	"	"	"	"
L23	3CH6	"	H	H	30	"	"	"	"	"	"	"
L24	3CH7	"	H	H	31	"	"	"	"	"	"	"
L25	1EN	"	L	L	6	27	32, 39	7, 40	"	"	"	"
L26	2EN	"	L	L	19	39	27, 32	"	"	"	"	"
L27	3EN	"	L	L	12	32	27, 39	"	"	"	"	"

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TABLE XIV. Switching time (t_{OFF}) test.

Test condition	Parameter	Switch position S2	Switch position S1	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
M1	1CH0 Switching time (t_{OFF})	L	L	6	7, 40	27, 32, 39	Open	0.03	0.80	µs	14, 15
M2	1CH1 "	L	L	4	"	"	"	"	"	"	"
M3	1CH2 "	L	H	25	"	"	"	"	"	"	"
M4	1CH3 "	L	H	5	"	"	"	"	"	"	"
M5	1CH4 "	H	L	23	"	"	"	"	"	"	"
M6	1CH5 "	H	L	2	"	"	"	"	"	"	"
M7	1CH6 "	H	H	24	"	"	"	"	"	"	"
M8	1CH7 "	H	H	3	"	"	"	"	"	"	"
M9	2CH0 "	L	L	19	"	"	"	"	"	"	"
M10	2CH1 "	L	L	37	"	"	"	"	"	"	"
M11	2CH2 "	L	H	17	"	"	"	"	"	"	"
M12	2CH3 "	L	H	38	"	"	"	"	"	"	"
M13	2CH4 "	H	L	35	"	"	"	"	"	"	"
M14	2CH5 "	H	L	15	"	"	"	"	"	"	"
M15	2CH6 "	H	H	36	"	"	"	"	"	"	"
M16	2CH7 "	H	H	16	"	"	"	"	"	"	"
M17	3CH0 "	L	L	12	"	"	"	"	"	"	"
M18	3CH1 "	L	L	33	"	"	"	"	"	"	"
M19	3CH2 "	L	H	13	"	"	"	"	"	"	"
M20	3CH3 "	L	H	34	"	"	"	"	"	"	"
M21	3CH4 "	H	L	28	"	"	"	"	"	"	"
M22	3CH5 "	H	L	29	"	"	"	"	"	"	"
M23	3CH6 "	H	H	30	"	"	"	"	"	"	"
M24	3CH7 "	H	H	31	" "	"	"	"	"	"	"
M25	1EN "	L	L	6	27	32, 39	7,40	"	"	"	"
M26	2EN "	L	L	19	39	27, 32	"	"	"	"	"
M27	3EN "	L	L	12	32	27, 39	"	"	"	"	"

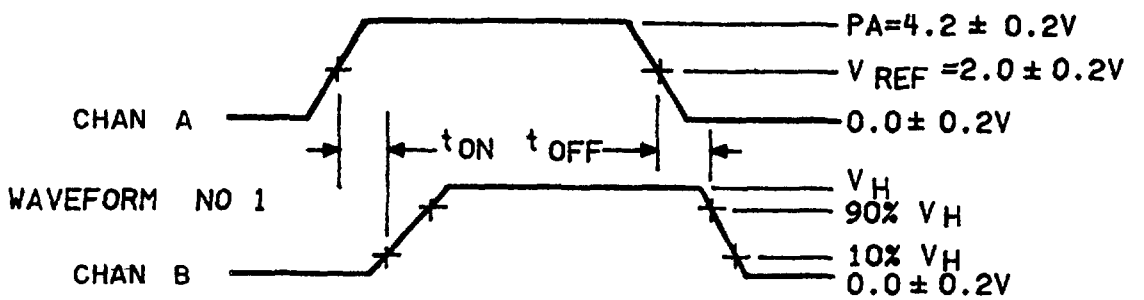
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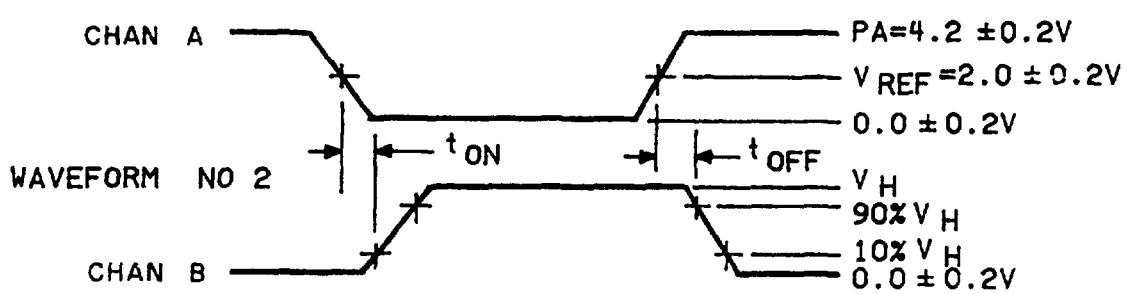


NOTES:
1. R1 = 51.1Ω ±5%, 1 W.
2. R2 = 2 kΩ ±1%, 1/8 W.
3. S1 = S2 = SPDT switch.
4. Item J01 setup:
   Frequency = 100 kHz ±10%.
   Duty cycle = 50% ±10%
   Down level = 0.0 ±0.2 V.
   Up level = 4.2 ±0.2 V.
   $t_{TLH} = t_{THL}$ = 30 ns maximum.

FIGURE 14. Switching time (t_{ON} and t_{OFF}) test circuit for tables XIII and XIV.

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NOTES:
1. Waveform number 1 applies to test conditions L2-L24 (even), M2-M24 (even), L25-L27 and M25-M27.
2. Waveform number 2 applies to test conditions L1-L23 (odd) and M1-M23 (odd).

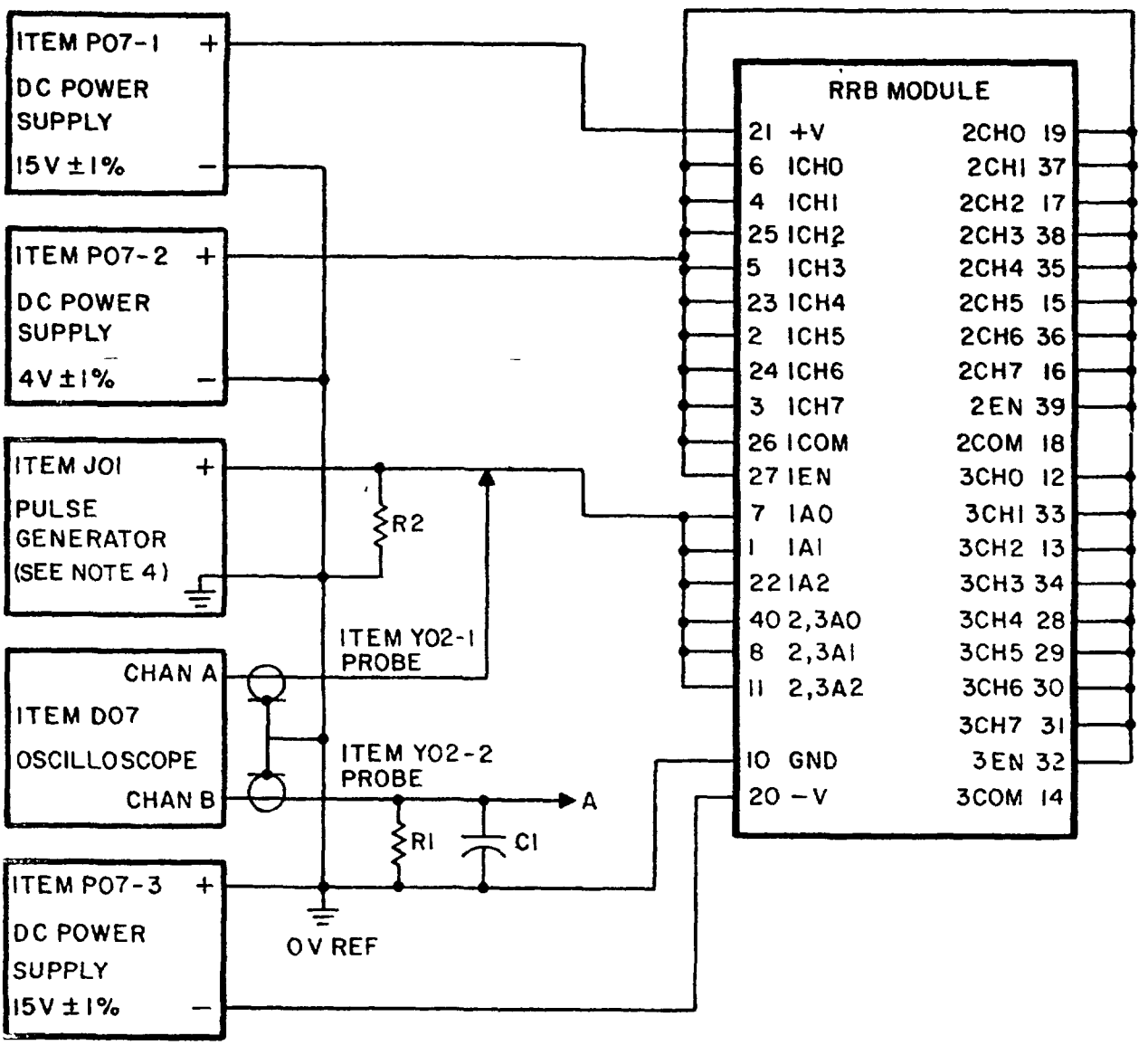
FIGURE 15. Switching time (t_{ON} and t_{OFF}) test circuit waveforms for tables XIII and XIV.

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TABLE XV. Break-before-make delay test.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
N1	1COM	Break-before-make delay	26	30	500	ns	16, 17
N2	1COM	"	26	"	"	"	"
N3	2COM	"	18	"	"	"	"
N4	2COM	"	18	"	"	"	"
N5	3COM	"	14	"	"	"	"
N6	3COM	"	14	"	"	"	"

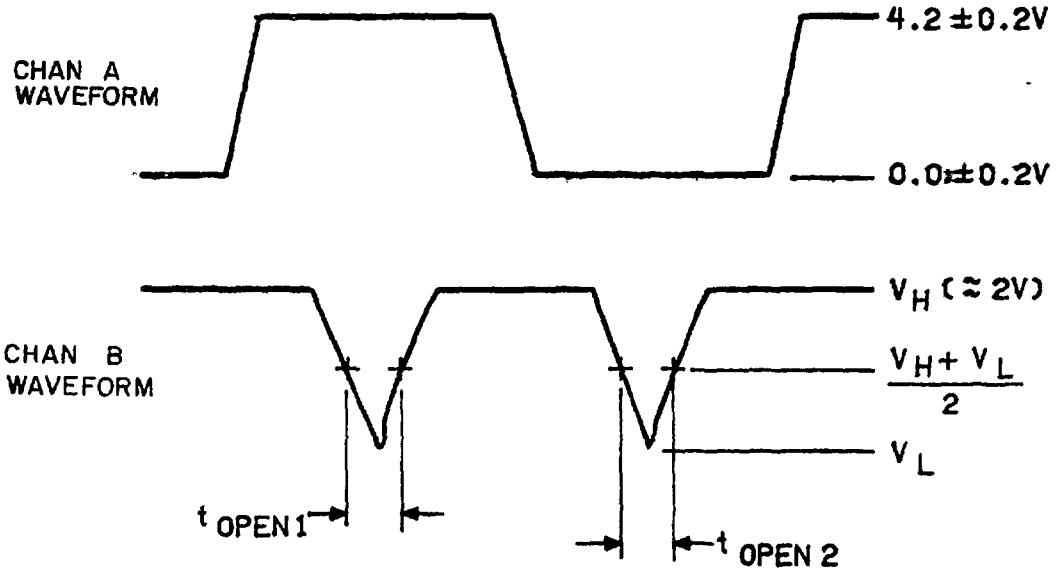
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NOTES:
1. R1 = 1 kΩ ±1%, 1/8 W.
2. R2 = 51.1Ω ±5%, 1 W.
3. C1 = 15 pF ±20%, 20 V dc.
4. Item J01 setup:
   Frequency = 100 kHz ±10%.
   Duty cycle = 50% ±10%.
   Down level = 0.0 ±0.2 V.
   Up level = 4.2 ±0.2 V.
   $t_{TLH} = t_{THL}$ = 30 ns maximum.

FIGURE 16. Break-before-make delay test circuit for table XV.

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NOTE: Measure t_{OPEN1} for test conditions N1, N3, and N5, and t_{OPEN2} for N2, N4, and N6.

FIGURE 17. Break-before-make delay test circuit waveforms for table XV.

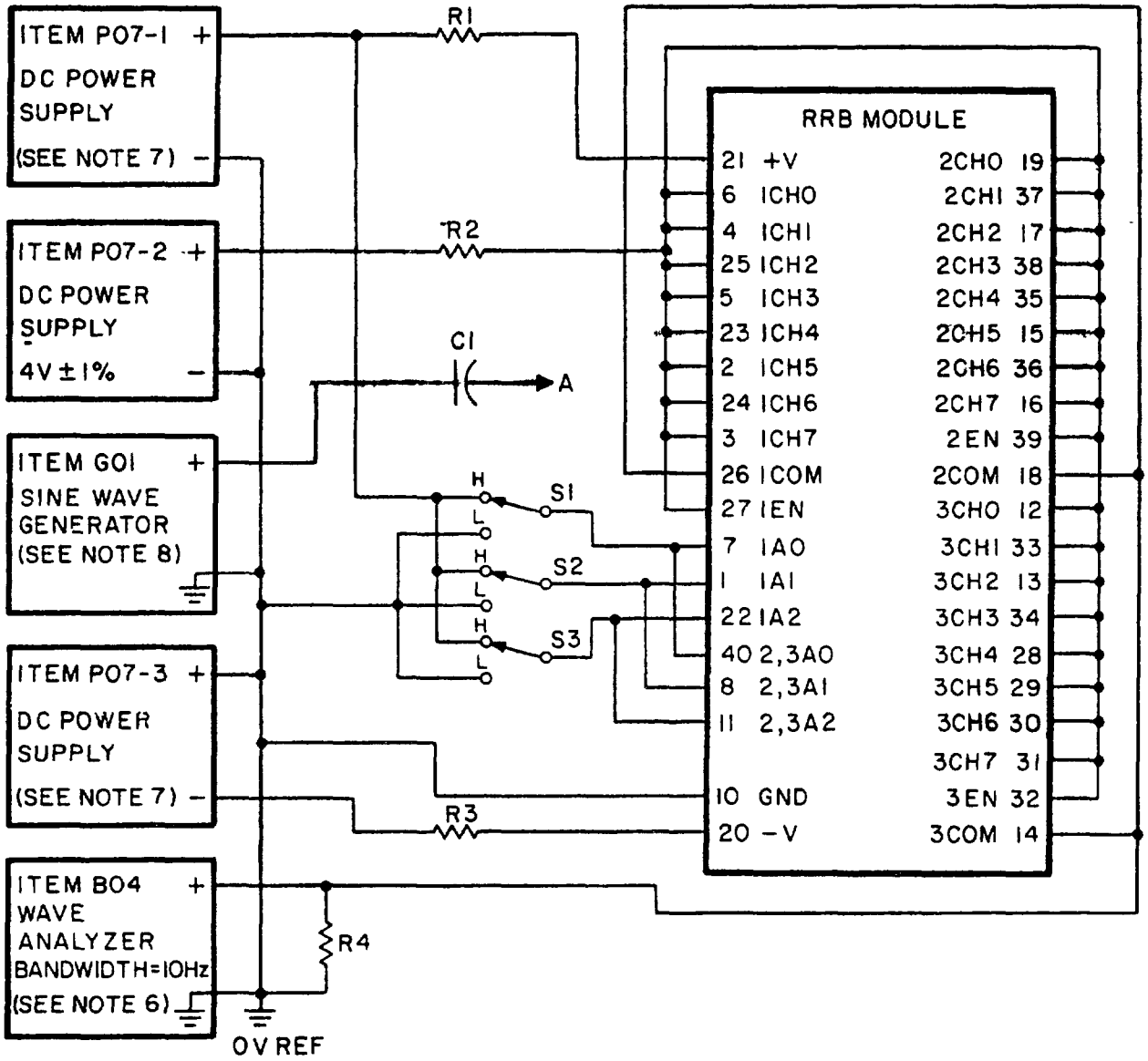
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*TABLE XVI. Power supply rejection ratio check test.

Test condition	Parameter	Switch position			Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		S3	S2	S1	A to pin	Min	Max		
P1	Power supply rejection ratio check 1/	L	L	L	21	0	700	µV rms	18
P2	"	L	L	H	"	"	"	"	"
P3	"	L	H	L	"	"	"	"	"
P4	"	L	H	H	"	"	"	"	"
P5	"	H	L	L	"	"	"	"	"
P6	"	H	L	H	"	"	"	"	"
P7	"	H	H	L	"	"	"	"	"
P8	"	H	H	H	"	"	"	"	"
P9	"	L	L	L	20	"	"	"	"
P10	"	L	L	H	"	"	"	"	"
P11	"	L	H	L	"	"	"	"	"
P12	"	L	H	H	"	"	"	"	"
P13	"	H	L	L	"	"	"	"	"
P14	"	H	L	H	"	"	"	"	"
P15	"	H	H	L	"	"	"	"	"
P16	"	H	H	H	"	"	"	"	"

1/ PSRR (table I) = 20 log $\frac{\text{Item B04 reading}}{700\ \mu\text{V rms}}$ = dB

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NOTES:
1. R1 = R3 = 1 kΩ ±1%, 1/8 W.
2. R2 = 51.1Ω ±5%, 1/8 W.
3. R4 = 100 kΩ ±1%, 1/8 W.
4. C1 = 1 µF ±10%, 20 V dc.
5. S1-S3 = SPDT switch.
6. Tune item B04 to 800 Hz and fine tune for maximum deflection.
7. Adjust item P07-1 to obtain 15 V ±1% at pin 21. Adjust item P07-3 to obtain -15 V ±1% at pin 20.
8. Adjust item G01 to obtain 0.7 V rms ±5%, 800 Hz ±1% at pin 21 for test conditions P1-P8 and pin 20 for test conditions P9-P16.

FIGURE 18. Power supply rejection ratio check test circuit for table XVI.

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TABLE XVII. Output crosstalk check test.

Test condition	Parameter	Switch position			Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
		S3	S2	S1	A to pin	B to pin 2/ except	C to pin	D to pin	E to pin	F to pin	Min	Max		
Q1	Output crosstalk check 1/	L	L	L	32, 39	No exception	26	C	Open	27	0	111	mV	19
Q2	"	L	L	L	27, 32	"	18	"	"	39	"	"	"	"
Q3	"	L	L	L	27, 39	"	14	"	"	32	"	"	"	"
Q4	"	L	L	L	39, 32	"	26	Open	C	27	0.0	11.1	"	"
Q5	"	L	L	L	27, 32	"	18	"	"	39	"	"	"	"
Q6	"	L	L	L	27, 39	"	14	"	"	32	"	"	"	"
Q7	"	L	L	L	27, 32 39	6	26	"	"	Open	"	"	"	"
Q8	"	L	L	H	"	4	"	"	"	"	"	"	"	"
Q9	"	L	H	L	"	25	"	"	"	"	"	"	"	"
Q10	"	L	H	H	"	5	"	"	"	"	"	"	"	"
Q11	"	H	L	L	"	23	"	"	"	"	"	"	"	"
Q12	"	H	L	H	"	2	"	"	"	"	"	"	"	"
Q13	"	H	H	L	"	24	"	"	"	"	"	"	"	"
Q14	"	H	H	H	"	3	"	"	"	"	"	"	"	"
Q15	"	L	L	L	"	19	18	"	"	"	"	"	"	"
Q16	"	L	L	H	"	37	"	"	"	"	"	"	"	"
Q17	"	L	H	L	"	17	"	"	"	"	"	"	"	"

See footnotes at end of table.

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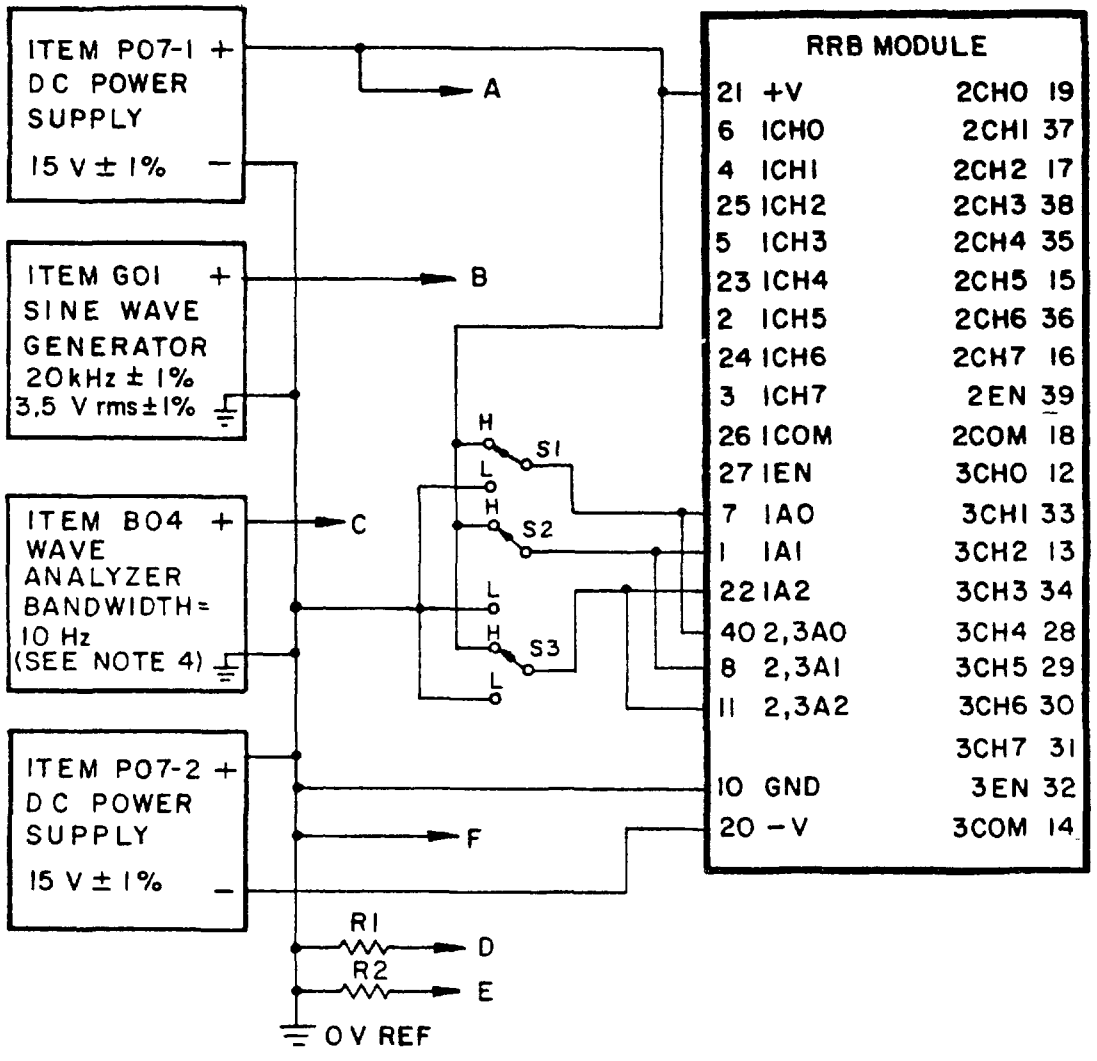
TABLE XVII. Output crosstalk check test - Continued.

Test condition	Parameter	Switch position			Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
		S3	S2	S1	A to pin	B to pin 2/ except	C to pin	D to pin	E to pin	F to pin	Min	Max		
Q18	Output crosstalk check 1/	L	H	H	27, 32 39	38	18	Open	C	Open	0.0	11.1	mV	19
Q19	"	H	L	L	"	35	"	"	"	"	"	"	"	"
Q20	"	H	L	H	"	15	"	"	"	"	"	"	"	"
Q21	"	H	H	L	"	36	"	"	"	"	"	"	"	"
Q22	"	H	H	H	"	16	"	"	"	"	"	"	"	"
Q23	"	L	L	L	"	12	14	"	"	"	"	"	"	"
Q24	"	L	L	H	"	33	"	"	"	"	"	"	"	"
Q25	"	L	H	L	"	13	"	"	"	"	"	"	"	"
Q26	"	L	H	H	"	34	"	"	"	"	"	"	"	"
Q27	"	H	L	L	"	28	"	"	"	"	"	"	"	"
Q28	"	H	L	H	"	29	"	"	"	"	"	"	"	"
Q29	"	H	H	L	"	30	"	"	"	"	"	"	"	"
Q30	"	H	H	H	"	31	"	"	"	"	"	"	"	"

1/ Output crosstalk (table I) = 20 log $\frac{\text{Item B04 reading}}{3.5\ \text{V rms}}$ = dB

2/ Connect point B to the following pins, except as listed above:
2, 3, 4, 5, 6, 12, 13, 15, 16, 17, 19, 23, 24, 25, 28, 29, 30, 31, 33, 34, 35, 36, 37, 38.

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NOTES:
1. R1 = 20 kΩ ±5%, 1/8 W.
2. R2 = 1 kΩ ±1%, 1/8 W.
3. S1 - S3 = SPDT switch.
4. Tune item B04 to 20 kHz and fine tune for maximum deflection.

FIGURE 19. Output crosstalk check test circuit for table XVII.

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*TABLE XVIII. Power supply filter capacitance, input and output capacitance test.

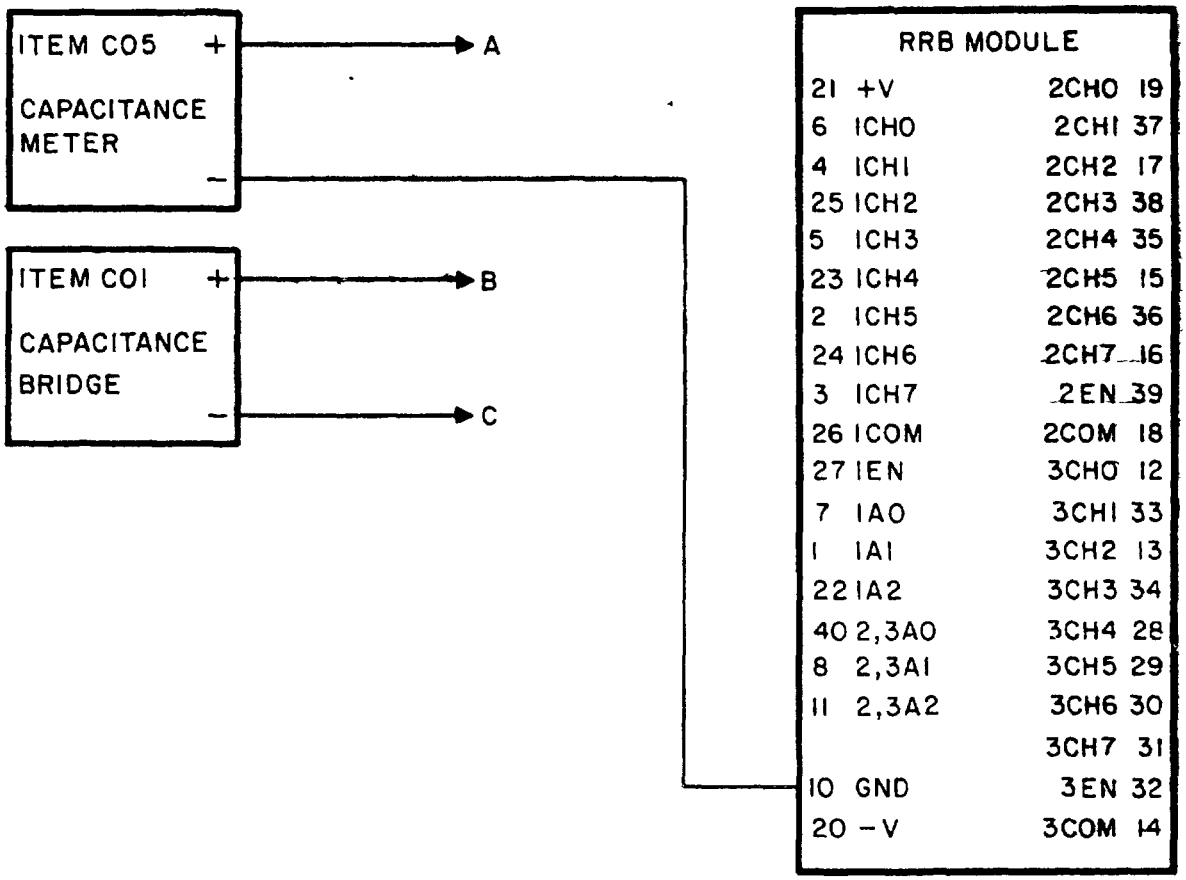
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R1	Power supply filter capacitance	Open	21	10	0.75	1.25	µF	20
R2	Power supply filter capacitance	Open	10	20	0.75	1.25	µF	"
S1	Input capacitance	40	Open	Open	2	30	pF	"
S2	"	8	"	"	"	"	"	"
S3	"	11	"	"	"	"	"	"
S4	"	7	"	"	"	15	"	"
S5	"	1	"	"	"	"	"	"
S6	"	22	"	"	"	"	"	"
S7	"	27	"	"	"	"	"	"
S8	"	39	"	"	"	"	"	"
S9	"	32	"	"	"	"	"	"
S10	"	6	"	"	"	"	"	"
S11	"	4	"	"	"	"	"	"
S12	"	25	"	"	"	"	"	"
S13	"	5	"	"	"	"	"	"
S14	"	23	"	"	"	"	"	"
S15	"	2	"	"	"	"	"	"
S16	"	24	"	"	"	"	"	"
S17	"	3	"	"	"	"	"	"

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*TABLE XVIII. Power supply filter capacitance, input and output capacitance test. - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S18	Input capacitance	19	Open	Open	2	15	pF	20
S19	"	37	"	"	"	"	"	"
S20	"	17	"	"	"	"	"	"
S21	"	38	"	"	"	"	"	"
S22	"	35	"	"	"	"	"	"
S23	"	15	"	"	"	"	"	"
S24	"	36	"	"	"	"	"	"
S25	"	16	"	"	"	"	"	"
S26	"	12	"	"	"	"	"	"
S27	"	33	"	"	"	"	"	"
S28	"	13	"	"	"	"	"	"
S29	"	34	"	"	"	"	"	"
S30	"	28	"	"	"	"	"	"
S31	"	29	"	"	"	"	"	"
S32	"	30	"	"	"	"	"	"
S33	"	31	"	"	"	"	"	"
T1	Output capacitance	26	"	"	"	50	"	"
T2	"	18	"	"	"	"	"	"
T3	"	14	"	"	"	"	"	"

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NOTES:
1. Item C01 setup:
   Frequency = 120 Hz.
   V_{pk} = 0.3 V maximum.
2. Null out the effects of the test leads prior to connecting the module.
3. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.

FIGURE 20. Power supply filter capacitance, input and output capacitance test circuit for table XVIII.

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